Marketing Analytics

Regression and Marketing Interpretation

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Session 04

Competitors and Store Sales

- Stores in markets with more competitors can have higher sales.
- Demand attracts firms and also raises sales.
- The managerial question concerns the effect of an additional competitor.

A Conditional Mean

Relate store sales to observed marketing conditions.

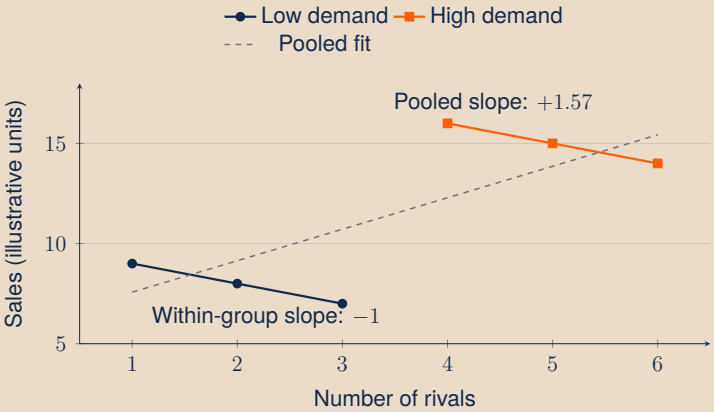
$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + u_i$$

The coefficient on X_i describes its linear association holding included Z_i fixed.

Ordinary Least Squares

- OLS chooses coefficients to minimize squared residuals.
- A residual is observed sales minus fitted sales.
- The model summarizes the comparison supplied by the data.

Omitted Demand Reverses the Sign



Synthetic markets hold the underlying competitive mechanism fixed.

Omitted-Variable Bias

Suppose demand Z affects sales and is omitted from a regression on X.

$$\widetilde{\beta}_1 = \beta_1 + \beta_2 \frac{\mathrm{Cov}(X, Z)}{\mathrm{Var}(X)}$$

The omitted association can offset or reverse the coefficient of interest.

Price and Sales Units

- A level coefficient is sales units per dollar.
- A log-log price coefficient is an elasticity under the model.
- The exact change for a log outcome uses an exponential transformation.

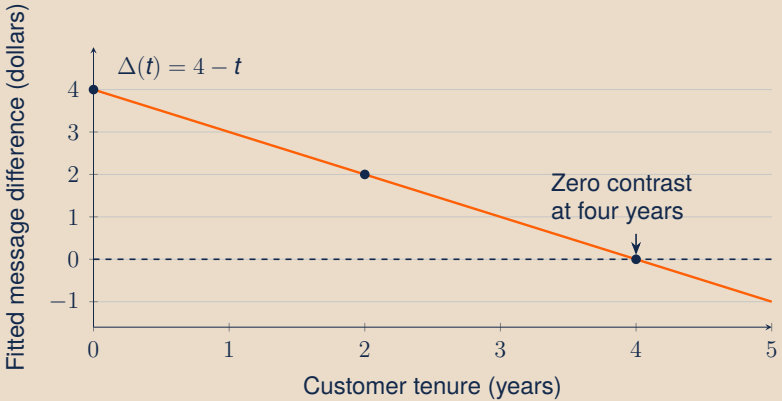
Interactions

Allow a message association to vary with account tenure T.

$$Y = \beta_0 + \beta_1 M + \beta_2 T + \beta_3 MT + u$$

The message comparison at tenure T is $\beta_1 + \beta_3 T$.

A Comparison Changes With Tenure



Synthetic message-response model. Tenure determines the conditional comparison.

Uncertainty and Practical Importance

- Report effect size in managerial units.
- Cluster at the unit that shares outcome disturbances or assignment.
- Compare the interval with a decision-relevant magnitude.

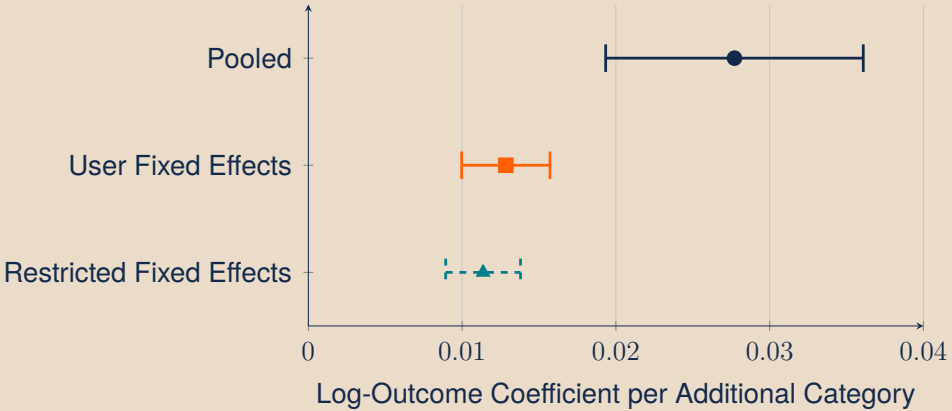
Panel Comparisons

- Customer effects absorb stable customer characteristics.
- Time effects absorb shocks shared within a period.
- Time-varying confounding can remain.

mHealth: The Next Calendar Day

- Relate recorded category breadth to next-day recorded calories.
- Match exact adjacent calendar dates.
- State the included users, days, and calorie restrictions.

Screening and the Estimated Association



mHealth estimates retain the stated outcome timing and sample restrictions.

A Hand-Checkable Regression

Synthetic sales follow a linear relationship.

```
x = [1, 2, 3, 4]
y = [3, 5, 7, 9]
mx, my = sum(x) / 4, sum(y) / 4
b = sum((a-mx)*(v-my) for a,v in zip(x,y))
b /= sum((a-mx)**2 for a in x)
a = my - b * mx
assert (a, b) == (1.0, 2.0)
```

The intercept is 1 and the slope is 2 sales units per unit of x.

Coding Exercise: Regression Checks

- Generate a model after defining the outcome and information set.
- Verify the coefficient units and one fitted value manually.
- Compare pooled and customer-effect specifications.
- Inspect residuals, outliers, and sample changes.

Prices Respond to Demand

- Managers can raise prices when demand is unusually strong.
- An observed price-sales regression can mix demand and pricing choices.
- An instrument needs relevance and a defensible exclusion restriction.

Evidence for an Advertised Claim

- A claim that a product cuts costs needs a suitable comparison.
- An association can reflect selection and omitted demand.
- The evidence should support the express and implied claim before publication.

Regression Discussion

- Which comparison supports the coefficient?
- What variable could reverse its interpretation?
- Would the recommended action change across the reported interval?